

$\therefore x$ is an irrational no. and y is a rational no.

By defn 2, there exists integers a and b where $b \neq 0$, $y = \frac{a}{b}$

By defn 2, there exists integers a_1 and b_1 where $b_1 \neq 0$,

$$x + y = \frac{a_1}{b_1}$$

$$\therefore x = (x + y) - y = \frac{a_1}{b_1} - \frac{a}{b} = \frac{a_1 b - a b_1}{b_1 b}$$

It's easy to see that $(a_1 b - a b_1)$ and $b_1 b$ are integers. $\because b \neq 0$ and $b_1 \neq 0 \therefore b_1 b \neq 0$

By defn 2, x is a rational no.

This contradicts the fact that x is an irrational no. $\therefore \neg r$ is False. By defn. of AND, $(p \wedge q) \wedge \neg r$ is False.

By defn. of conditional, $((p \wedge q) \wedge \neg r) \rightarrow F$ is True. $\therefore (p \wedge q) \rightarrow \neg r$ is True.

$\because (p \wedge q)$ is True, r is True (Modus ponens) $\therefore xy$ is an irrational no. $\square$

10) Use a direct proof to show that the product of two rational numbers is rational.

Soln: $p: x$ is a rational number $q: y$ is a rational no.

$r: xy$ is rational

We shall prove $(p \wedge q) \rightarrow r$ is True by a direct proof.

Let $(p \wedge q)$ be True. By defn. of AND, p is True and q is True.

$\therefore x$ and y are arbitrary rational numbers.

By defn. 2, there exists integers a, b where $b \neq 0$ s.t. $x = \frac{a}{b}$

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It's easy to see that ac and bd are integers.

$\because b \neq 0$ and $d \neq 0$, $\therefore bd \neq 0$.

By defn 2, xy is a rational no. $\therefore r$ is True.

By defn. of conditional, $(p \wedge q) \rightarrow r$ is True $\square$